SDR · SATELLITE · 10 GHZ

Getting on QO-100 with sdroxide

Step-by-step setup for working the QO-100 (Es'hail-2) geostationary amateur satellite with an **ADALM-Pluto** and a **Ku-band LNB**. The 10 GHz downlink comes down through the LNB; the 2.4 GHz uplink leaves the Pluto directly. The settings below are done once — after that it is just `SAT` and beacon sync.

REQUIREMENTS

- **ADALM-Pluto** (or a LibreSDR / ANTSDR-class AD936x board).
- A universal **Ku-band LNB** with a **9750 MHz** local oscillator — the standard on QO-100 stations.
- A dish, the coax from the LNB, and a 2.4 GHz uplink antenna.
- **sdroxide** installed — the Pluto driver is built in, nothing else to install.

```
Downlink:  10,489.750 MHz  →  LNB (LO 9750 MHz)  →  739.750 MHz  →  Pluto
Uplink:     2,400.x MHz  ←  Pluto (direct, no conversion)
```

Bold `like this` is a button or field name exactly as it appears on screen; `this style` is a menu path or a value you type.

PART A

Settings

Done once and saved. Everything is in the `Settings` window, on the `General` and `Radio` tabs.

STEP 01

Start sdroxide

Open the program. The main window has the control bar across the top and the panadapter and waterfall below it.

STEP 02

Open the settings

Press `SETTINGS` on the top control bar.

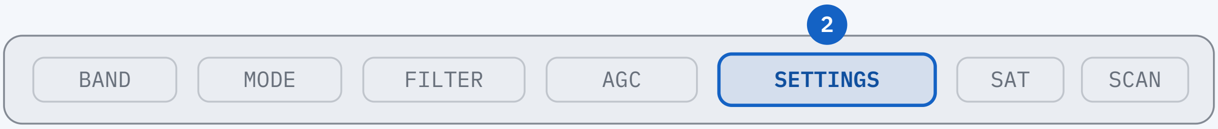

Figure 1 — The top control bar. 2: SETTINGS.

STEP 03

Callsign and locator

On the `General` tab, type your callsign into `Callsign` and your Maidenhead grid (e.g. `KN41GG`) into `Locator`. The satellite lock and the world map use this locator — `LOCK ON` will not work without it.

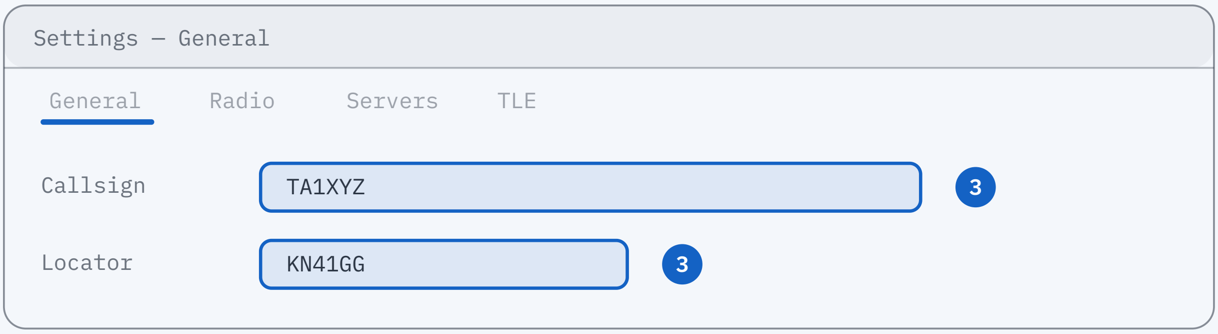

Figure 2 — The General tab. 3: Callsign + Locator.

STEP 04

Radio tab — device: PlutoSDR

Switch to the `Radio` tab. Pick `PlutoSDR` in the interface selector. A Pluto is a network device even on a USB cable: plugging it in creates a network adapter, not a serial port.

The next five fields are on this same tab — all of them in the drawing below:

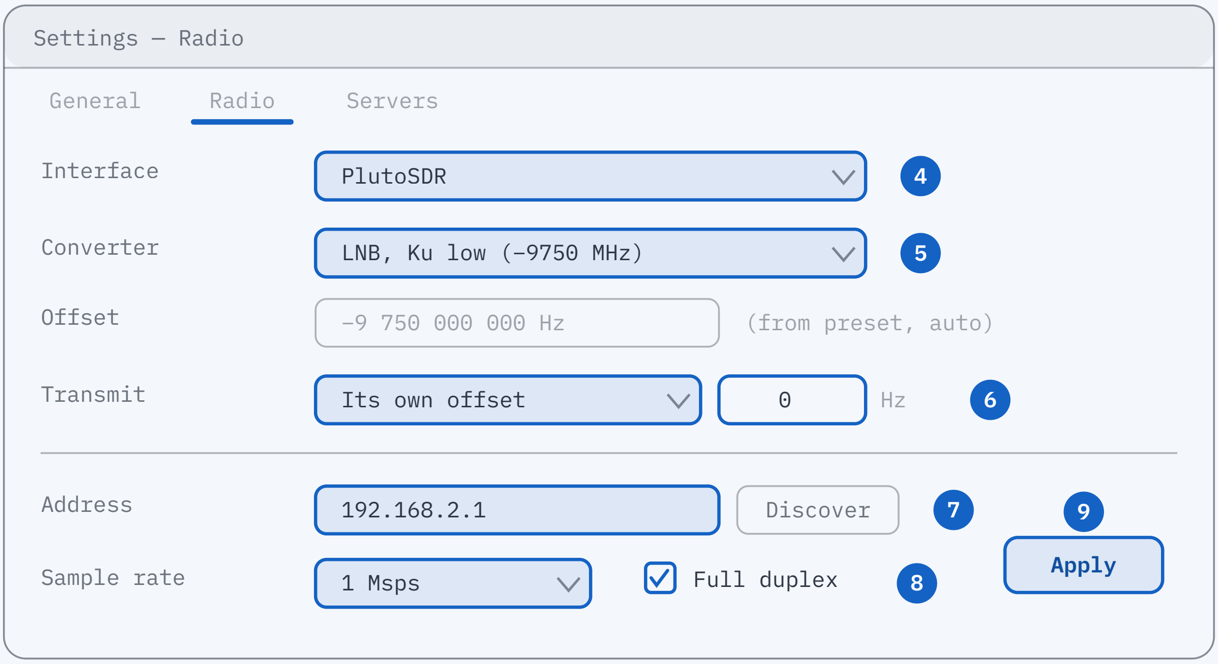

Figure 3 — The Radio tab. The fields that change for QO-100 are marked 4–9.

STEP 05

Converter: LNB, Ku low (−9750 MHz)

In the `Converter` dropdown, pick the `LNB, Ku low (−9750 MHz)` preset — because our LNBs have a 9750 MHz local oscillator. This fills in the `−9 750 000 000 Hz` offset below automatically, so the 10,489.750 MHz beacon lands near the right place on the dial.

STEP 06

Transmit row: Its own offset = 0

In the `Transmit` row next to the converter offset, choose `Its own offset` and type `0` into the Hz box beside it.

There is no need to enter a TX figure like 8085 MHz the way we used to. This `0` means "the downlink comes through the LNB, but the 2.4 GHz uplink leaves the radio directly — the transmit side is not converted". If you later decide your TX frequency is off, you correct it from this box (in Hz, same sign rule as the receive offset).

STEP 07

Pluto address

In the connection section below, type the Pluto's IP into `Address` or let `Discover` find it. Over USB, `192.168.2.1` is enough. On a network, use the Pluto's address on that network. `ip:192.168.2.1` is also accepted; an address starting with `usb:` is refused, because this backend reaches the radio over the network the USB cable already provides.

STEP 08

Sample rate and Full duplex

Set `Sample rate` to `1 Msps`. Tick `Full duplex` below it — so you keep hearing your own downlink even while transmitting.

> **NOTE**
>
> `Full duplex` only works over real Ethernet: a Pluto or LibreSDR behind a gigabit adapter. Over USB the station is half-duplex and you will not hear your own downlink during an over. Also, a stock Pluto cannot go below about 2.084 Msps; a lower value is rounded up and the connection message says so.

STEP 09

Apply and close

Press `Apply` at the bottom and close the `Settings` window. Settings take effect on Apply / reconnect.

PART B

Locking onto the satellite

Done every time you work the satellite.

STEP 10

Open the SAT window

Press `SAT` in the System box. The button glows green while a lock is running; the correction keeps being applied even with the window closed.

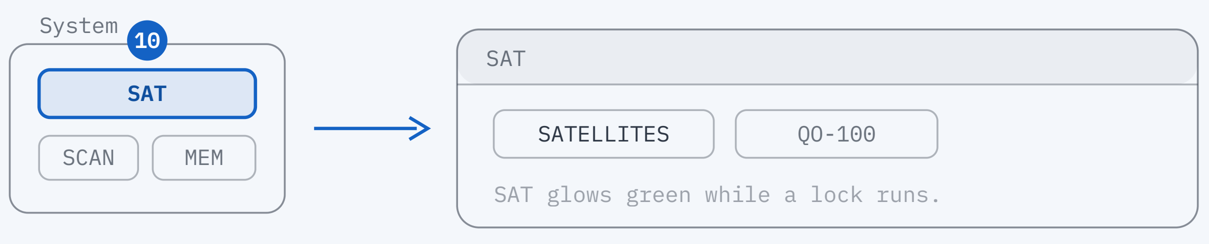

Figure 4 — 10: SAT in the System box. The window has two tabs: SATELLITES and QO-100.

STEP 11

Pick QO-100

In the `SATELLITES` tab, type `QO-100` into the search box and pick it from the list. Its published links (narrowband transponder, beacon) appear below.

STEP 12

LOCK ON

Press `LOCK ON` at the bottom. QO-100 is geostationary, so it needs no Doppler correction; the transponder mapping derives the 2.4 GHz uplink from wherever you set the downlink dial.

Figure 5 — The SATELLITES tab. 11: the QO-100 row · 12: LOCK ON.

PART C

Beacon sync

Optional but recommended. If your LNB is not very stable, the QO-100 beacon sync plugin developed by the ATÖLYE group corrects LNB drift continuously.

STEP 13

Switch to the QO-100 tab

Switch to the `QO-100` tab — the second tab of the SAT window. This plugin measures where the 10,489.750 MHz narrowband beacon really is, corrects the converter/LNB offset so the dial and the signal agree, and keeps correcting it as the LNB warms up and drifts.

> **PREREQUISITE**
>
> The plugin corrects a *residual* error of a few kHz to tens of kHz — it does not guess from scratch. The `Settings › Radio` converter offset has to be roughly right first (Step 05) so the beacon lands somewhere near 10,489.750 MHz.

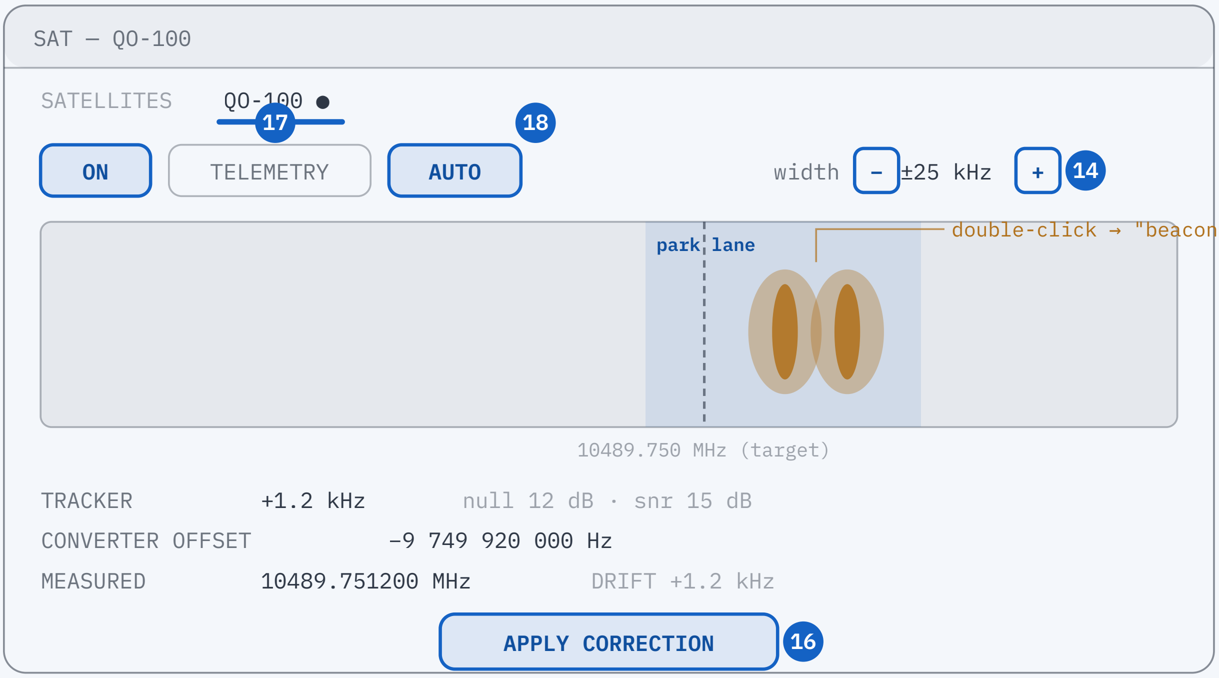

Figure 6 — The QO-100 tab. 14: width · 16: APPLY CORRECTION · 17: ON + AUTO · 18: TELEMETRY.

STEP 14

See the beacon in the strip

If you cannot see the beacon in the strip as a band / two symmetric lobes, step the `width` value up with `+` (±5 kHz steps, up to ±50 kHz). Widen only until the two lobes and the null between them are clear — then bring it back toward ±5 kHz.

STEP 15

Still nothing: nudge 9750 by hand

Nudge the 9750 MHz converter offset in `Settings › Radio` to an approximate value in the right direction and press `Apply`; repeat until the beacon is inside the strip. The plugin takes it from there.

STEP 16

Simple (one-shot) correction

Double-click the middle of the beacon in the strip — that plants a "the beacon is here" mark (the two lobes, centred on the null). Then press `APPLY CORRECTION` at the bottom. The receiver reopens on the corrected offset and the beacon jumps toward the centre. Two passes — one wide (coarse), one narrow (fine) — get it within a few hundred Hz.

STEP 17

Automatic, continuous correction

For continuous correction, press `ON` (the spectral tracker starts — it only *measures*, it changes nothing) and then `AUTO` (the loop closes: every clean, steady measurement is applied as a slow, rate-limited nudge to the offset).

AUTO corrects LNB drift in the background. A single noisy reading never yanks the receiver; it takes a run of agreeing measurements to move the offset. No correction is made *while you are transmitting* (each one reopens the receiver, which would cut off your over) — a held-back correction goes out as soon as the over ends. AUTO also watches its own work and switches itself off if the offset it writes does not move the beacon, or if it has gone further than any LNB could plausibly be out by, with a notice saying which.

STEP 18

Telemetry (optional)

To decode the beacon's telemetry, press `TELEMETRY` — it runs the AO-40 frame decoders and shows the beacon's own status text when it locks. It is not needed for calibration; it is an independent check.

STEP 19

Leave it running in the background

When you are done, minimise this window and keep it in a corner of the screen (closing it is fine too). The LNB frequency correction keeps going with the window closed; the `SAT` chip stays lit and the QO-100 tab keeps its dot.

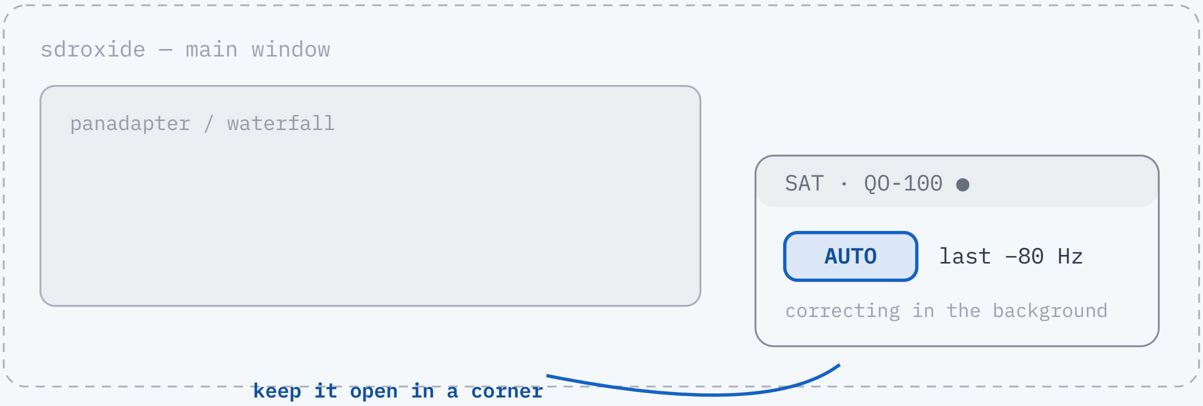

Figure 7 — The panel minimised to a corner; AUTO keeps correcting even when it is closed.

Good QSOs, and 73!

This guide is based on sdroxide's QO-100 beacon plugin and its Pluto backend. For detail, see §2.21 (QO-100 beacon plugin) and §6.2.7 (PlutoSDR) in the program's `docs/USER_MANUAL.md`.